

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

17% Global Traffic Growth and Starlink's 3.3x Surge

- Global Internet traffic grew by 17% in 2024, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.
- SpaceX Starlink traffic saw dramatic 3.3x growth during 2024, continuing to bring satellite Internet connectivity to previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

- Google maintained its position as the most popular Internet service overall for the second consecutive year.
- Generative AI continued its rapid rise throughout 2024, while the Metaverse

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Post-Quantum Encryption Jumps from 2% to 13% of TLS

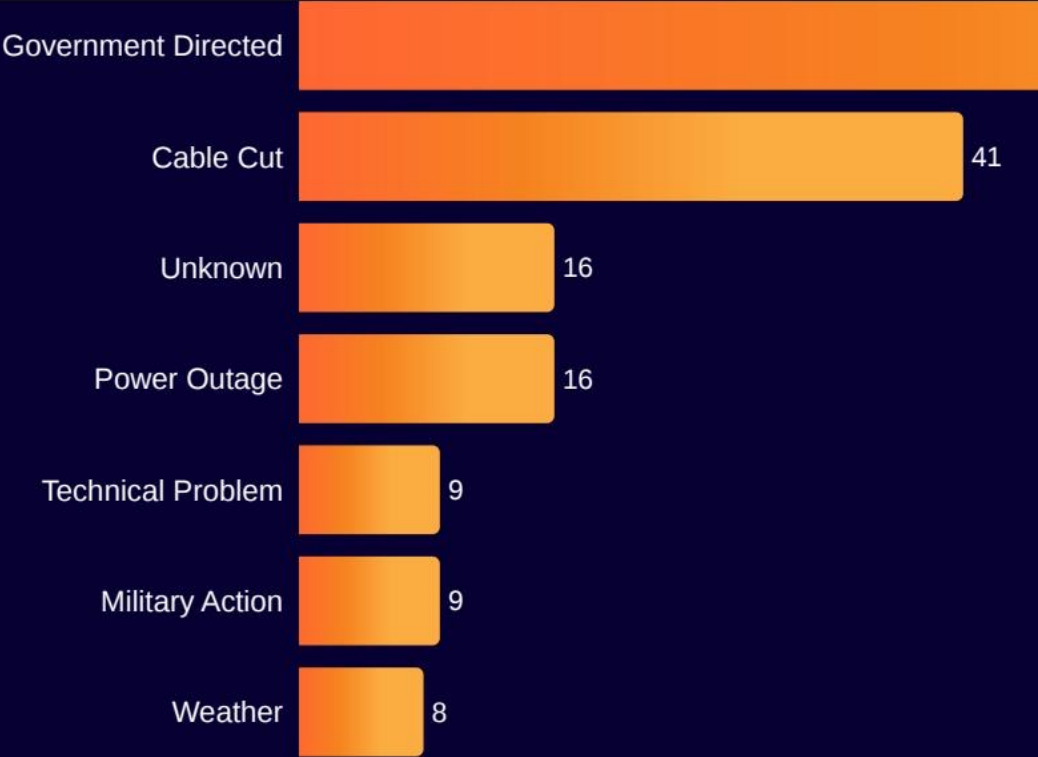
1.3

- Post-quantum encryption saw a dramatic surge, jumping from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.
- HTTP/3 adoption reached 21% of traffic.
- Go emerged as the most popular API client language with roughly 12% of automated API requests, followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).



225 Major Disruptions: Government Shutdowns Lead Outage Causes

- Cloudflare observed 225 major Internet disruptions globally in 2024.
- Government-directed shutdowns were the leading cause (116+), followed by cable cuts (41) and power outages (16).



Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

- 6.5% Mitigation Rate: Cloudflare mitigated 6.5% of all traffic it processed, reflecting the massive scale of ongoing cyber threats.
- Most Attacked Sector: The Gambling and Games sector remained the most attacked industry category.
- Log4j Persistence: The CVE-2021-44228 vulnerability continues to be actively targeted more than two years after initial disclosure.
- Bot Concentration: 68% of global bot traffic originated from just the top 10 countries.
- Email Security: 4.3% of processed emails were malicious; 43% of these contained deceptive links designed to trick recipients.

Five Takeaways Shaping the 2025 Internet Outlook

1. Unabated Traffic Growth: Global Internet traffic grew 17% year-over-year, with Starlink surging 3.3x to underserved regions.
2. Security-First Protocol Adoption: Post-quantum encryption leapt to 13% of TLS 1.3, highlighting swift industry action on future threats.
3. Connectivity Fragility: Government-directed shutdowns remain the dominant cause of Internet outages, outpacing physical infrastructure issues.
4. Persistent Exploitation: Foundational vulnerabilities like Log4j remain pervasive, requiring relentless zero-trust enforcement.
5. API Ecosystem Shifts: Go has surpassed other languages to dominate automated API traffic (12%), signaling changes in developer tooling.